

HIGH ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report reviews the single patent identified in the manifest for relevance to high-acrylonitrile nitrile butadiene rubber (NBR). The target material is a nitrile rubber copolymer enriched in acrylonitrile to increase polarity and oil resistance, typically achieved by controlling monomer feed ratio and polymerization conditions during emulsion or solution copolymerization. The patent analyzed here does not disclose a new high-ACN NBR polymerization recipe; instead, it addresses downstream metathesis degradation of nitrile rubber, including NBR, using ruthenium- or osmium-based catalyst systems combined with salts. The disclosed chemistry is relevant because it operates directly on nitrile rubber feedstocks and therefore intersects with NBR composition, molecular weight, and processability considerations that are important for high-ACN grades.

The patent landscape represented by this manifest is narrow and process-oriented rather than composition-oriented. The patent teaches that nitrile rubber can be subjected to metathesis in the presence of a catalyst system comprising a metathesis catalyst and one or more salts, with a salt-to-catalyst weight ratio from 0.01:1 to 10000:1. The examples focus on ring-closing metathesis model reactions using Grubbs II catalyst with LiBr or CsBr, demonstrating accelerated conversion relative to catalyst alone. The broader specification discusses metathetic degradation of nitrile rubber and the role of catalyst activity in reducing molecular weight before hydrogenation, but it does not provide a high-acrylonitrile copolymerization route or explicit acrylonitrile feed ratios.

Overall, the evidence indicates that the most important process themes for high-ACN NBR remain monomer composition control, polymerization temperature, and conversion control, but these are not directly disclosed in the patent manifest provided. The patent is therefore only indirectly relevant to high-acrylonitrile NBR synthesis, mainly as a post-polymerization processing technology for nitrile rubber feedstocks.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a metathesis process for reacting nitrile rubber in the presence of a catalyst system comprising a metathesis catalyst and one or more salts. The specification emphasizes improved catalyst activity for metathetic degradation of nitrile rubber, while the examples demonstrate salt-assisted ring-closing metathesis model reactions using Grubbs II catalyst. - Salt-to-metathesis catalyst weight ratio: 0.01:1 to 10000:1 — Claim 1 - Catalyst metal: osmium or ruthenium — Claim 1 - Example 7.01: 10 mg Grubbs II catalyst — Example 7 - Example 7.02: 10 mg Grubbs II catalyst — Example 7 - Example 7.02: 1 mg LiBr — Example 7 - Example 8.02: 1 mg CsBr — Example 8 - Example 7: 0.3 ml chlorobenzene — Example 7 - Example 7: 0.2 ml CDCl₃ — Example 7 - Example 7: 0.15 ml diethyl diallylmalonate — Example 7 - Example 7: room temperature — Example 7 - Example 8: room temperature — Example 8 - Example 7: 2 minutes before substrate addition — Example 7

Relevant Experimental Evidence - Example 7: LiBr addition accelerated ring-closing metathesis of diethyl diallylmalonate relative to catalyst alone. - Example 8: CsBr addition also accelerated metathesis, though the effect was less pronounced than in the LiBr case. - Example 7 and 8: Grubbs II catalyst was used in NMR-tube experiments at room temperature to compare salt-assisted and salt-free conversion profiles.

Technical Relevance This patent is relevant because it addresses metathesis processing of nitrile rubber feedstocks, which can be used to modify molecular weight and processability of NBR-derived materials. However, it does not disclose the synthesis of high-acrylonitrile NBR itself or any acrylonitrile-rich copolymerization conditions.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The available evidence does not disclose a direct high-acrylonitrile NBR polymerization patent; instead, it shows a metathesis-based post-polymerization process for nitrile rubber. The most concrete parameters are the salt-to-catalyst ratio of 0.01:1 to 10000:1, use of ruthenium or

osmium metathesis catalysts, and room-temperature salt-assisted model reactions with 10 mg Grubbs II catalyst and 1 mg LiBr or CsBr. For high-ACN NBR synthesis specifically, the manifest provides no explicit acrylonitrile feed ratios, emulsion recipe, or polymerization temperature window. Based on the broader target definition, the recommended synthesis focus remains controlled copolymerization of acrylonitrile and butadiene with tight monomer feed control, but that recommendation is not directly supported by this patent's examples.

5. REFERENCES

- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>